

1 →

$$x = d \times 2^m \quad 1 \leq d < 2 \quad -126 \leq m \leq 127$$

$$(1.a \dots a_{23})_2 \times 2^m$$

$$x_{left} = (1 \dots a_{23})_2 \times 2^m$$

$$x_{Right} = ((1 \dots a_{23})_2 + 2^{-13}) \times 2^m$$

$$x_L \leq x \leq x_R$$

→ Result

$$x = 2^5 + 2^{-16} + 2^{-19}$$

$$x = 2^5 (1 + 2^{-17} + 2^{-24})$$

$$x \rightarrow (1. \overset{15}{\underset{\uparrow}{00000000000000000001}} \overset{24}{\underset{\uparrow}{00000000000000000001}}) \cdot 2^5$$

$$x_{left} = (1. \overset{15}{\underset{\uparrow}{00000000000000000001}} \overset{24}{\underset{\uparrow}{00000000000000000000}}) \cdot 2^5$$

$$= 2^5 (1 + 2^{-17})$$

$$x_{Right} = (1. \overset{15}{\underset{\uparrow}{00000000000000000001}} \overset{24}{\underset{\uparrow}{00000000000000000001}}) \cdot 2^5$$

$$= \boxed{2^5 (1 + 2^{-17} + 2^{-23})}$$

$$4 \rightarrow f(x) = x^3 + 2x^2 - 5x - 6$$

$$x_0 = 1$$

$$\rightarrow f'(x) = 3x^2 + 4x - 5 \rightarrow x = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$\rightarrow x_1 = 1 - \frac{1+2-5-6}{3+4-5} = 1 - \frac{-8}{2} = \boxed{5}$$

$$\rightarrow x_2 = 5 - \frac{5^3 + 2 \cdot 5^2 - 5 \cdot 5 - 6}{3 \cdot 5^2 + 4 \cdot 5 - 5}$$

$$\rightarrow x_2 = 5 - \frac{144}{90} \rightarrow 5 - 1,6 = \boxed{3,4}$$

$$3 \rightarrow f(x) = x^3 + 2x^2 - 5x - 6$$

$$x_0 = 1 \quad x_1 = -1$$

$$f(x_0) = x^3 + 2x^2 - 5x - 6 \quad f(x_1) = x^3 + 2x^2 - 5x - 6$$

$$x_0 = \frac{1+2-5-6}{3-5-6} = -8 \quad f(-1) = -1 + 2 \cdot 1 - 5 - 6$$

$$-1 + 2 + 5 - 6$$

$$+1 - 1 = 0$$

$$\rightarrow x_{n+1} = x_n - \frac{f(x_n) \cdot (x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}$$

$$x_2 = -1 - 0 \cdot \frac{-1-1}{0-1-8} = -1 + 0 = \boxed{-1}$$

2 →

$$x = 0,643636895$$

$$y = 0,643631734$$

$$\left| \frac{(x+y) - (x+y)}{x+y} \right|$$

$$\boxed{x+y}$$

abre

$$x = 0,643677 \quad y = 0,643672$$

$$x+y = 1,2873449$$

$$100 \times \left| \frac{1,287348629 - 1,287341}{1,287348629} \right| = \text{Error}$$

$$x = 0,0002881912113$$

$$\boxed{= 0,00028}$$

$$\boxed{x-y}$$

$$100 \times \frac{(0,000000516) - (0,00000005)}{0,00000516}$$

= error

$$r. 3,119550474714703$$

$$\boxed{r. 3,1195}$$